17 Infrainguinal Reconstruction: Assessment and Reporting

Medical Translation Studio Test Group

CHAPTER SCOPE. This original test chapter imitates the structure of a surgical textbook. It contains invented records, measurements and references to challenge document extraction. It is not a clinical reference and contains no treatment recommendations.

KEY CONCEPTS
1. Preserve chapter hierarchy, table relationships and figure-caption links.
2. Distinguish a measured zero from missing, not assessed and absent.
3. Keep the source record intact when translating or editing its interpretation.

17.1 PRINCIPLES OF A STRUCTURED RECORD

A reconstruction record combines narrative description, a diagram, numerical observations and an account of how those observations were obtained. These components occupy different regions of a printed page, but they describe one logical episode. An extraction system should preserve both the text and the relationship between the parts.[1,2]

Anatomic description

The synthetic model names an inflow segment, a conduit and two outflow branches. The labels are identifiers for this fixture, not a statement about the suitability of any clinical reconstruction. Figure 17.1 shows the identifiers in a raster illustration; the caption remains ordinary searchable text.

A side designation must remain attached to its record. In Table 17.1, R and L are deliberate test values. Neither abbreviation may be inferred from a neighboring row. A blank field is also meaningful and must not be silently replaced with the previous nonblank value.

17.2 SOURCE AND OBSERVATION

The source is divided into three layers: what was observed, how it was labeled, and what was subsequently discussed. The layers may be adjacent in prose, yet they should remain distinguishable in a structured representation. A translation should retain the author's stated uncertainty instead of converting it into a positive finding.

Measurement vocabulary

For this chapter, a value is an invented scalar attached to a unit. A threshold symbol belongs to the value. Thus, ≤ 0.80 is not interchangeable with ≥ 0.80, and 0.05 is not interchangeable with 0.5. Numeric examples are presented solely to test extraction fidelity.

The terms *absent*, *not assessed* and *not visible* describe different states in the fixture. They must not collapse into a generic missing-value label. See the reporting examples on printed page 405 and the synthetic references on printed page 406.

[1] *All case identifiers, numerical observations and bibliography entries in this chapter are fictional. No patient data or published textbook passage is reproduced.*

17.2.1 Orientation and sequence

Reading order is a property of the chapter rather than the order of graphic objects in the PDF stream. Read the left column to its end, then the right column. The illustration below spans both columns and follows their shared discussion. Repeated running headers do not belong inside either paragraph.

A reconstruction record is deliberately printed with one reconstruc-
tion break in this paragraph. A semantic text output may join that line-end break, but an archival extraction should retain enough source information to explain the repair. A lexical hyphen in a term such as follow-up is different.

The tokens finding and flow use typographic ligatures. A normalized output may return finding and flow, provided it neither loses letters nor introduces spaces within the words.

17.2.2 Figure relationships

Panel A is an original diagram with five labels embedded as pixels. Panel B is a generated signal with arbitrary axes. Neither panel contains a diagnostic cutoff. The two panels together form one logical figure, and the caption applies to the composite rather than to two independent figures.[3]

Text inside an illustration may need a separate OCR step even when the surrounding page contains searchable text. A parser should not conclude that every word on a native PDF page is available through its text objects. Preserve the image before attempting any label extraction.

The figure number may be cited as Fig. 17.1 or Figure 17.1 in the body. Both references should resolve to the same figure entity. The labels Inflow and Conduit should not be inserted into the middle of a paragraph simply because their coordinates align with its lines.

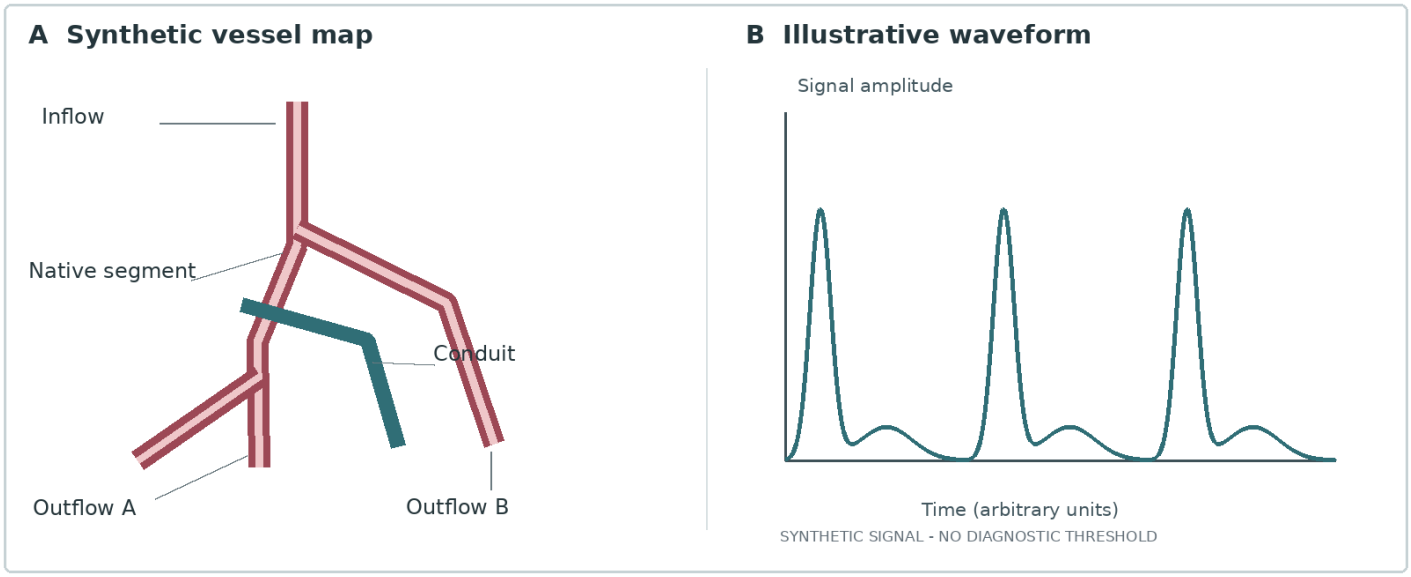


FIGURE 17.1 Original synthetic vessel map and signal trace. **A,** Inflow, conduit and outflow labels are embedded in the image. **B,** The waveform uses arbitrary units and carries no diagnostic threshold. Both panels belong to one figure.

> **INTERPRETATION NOTE**
> Image preservation and label recognition are separate tasks. An unsuccessful OCR result must not cause the underlying illustration to disappear.

17.3 SYNTHETIC OBSERVATIONS

Table 17.1 is intentionally continued across two physical pages. Its top header has merged column groups, and its first column contains vertical group labels. Repeated headers on the continuation page are furniture of the table, not new observations.[4]

TABLE 17.1 Invented reconstruction records (part 1 of 2)

Case identity			Synthetic measurements			Audit fields	
Group	Case ID	Side	Diameter (mm)	Flow index	Window (days)	State	Record note
A	T001	R	4.5	0.05	0	present	Baseline only
	T002	L	5.0	0.50	7	not assessed	No scalar value*
	T003	R	5.5	0.80	14	absent	Zero retained
B	T004	L	6.0	1.00	30	present	Repeat label
	T005	R	6.5	1.20	90	not visible	Not a negative result
	T006	L	7.0	1.50	180	present	Synthetic pair
C	T007	R	4.5	0.05	0	absent	Baseline only
	T008	L	5.0	0.50	7	present	No scalar value*
	T009	R	5.5	0.80	14	not assessed	Zero retained

** A missing measurement is not the number zero. Units and labels are test data only. Table continues on printed page 404.*

Grouped records

Groups A, B and C each contain three distinct cases. The group label spans the cases vertically, while the case identifier remains unique on every row. Flattening the first column should propagate only the group label and must not merge case identifiers or neighboring measurements.

The phrase not assessed is retained as text. It is not an instruction to invent a value. The symbol next to a note refers to the note below the table, not to the next row.

Continuation rules

The next page repeats the title, the two header rows and the units. A single logical table should be reconstructed from both fragments. It contains eighteen body rows, regardless of how often the header is printed.

A parser may retain physical fragments as children of the logical table. This is preferable to losing their page coordinates. The source-side order remains T001 through T018; sorting by a measurement would change the printed record.

TABLE 17.1 Invented reconstruction records (continued)

Case identity			Synthetic measurements			Audit fields	
Group	**Case ID**	**Side**	**Diameter (mm)**	**Flow index**	**Window (days)**	**State**	**Record note**
D	T010	L	6.0	1.00	30	present	Repeat label
	T011	R	6.5	1.20	90	absent	Not a negative result
	T012	L	7.0	1.50	180	present	Synthetic pair
E	T013	R	4.5	0.05	0	not visible	Baseline only
	T014	L	5.0	0.50	7	present	No scalar value*
	T015	R	5.5	0.80	14	not assessed	Zero retained
F	T016	L	6.0	1.00	30	present	Repeat label
	T017	R	6.5	1.20	90	absent	Not a negative result
	T018	L	7.0	1.50	180	present	Synthetic pair

** Header units apply to both fragments. The table ends here. No additional observation is contained in this note.*

17.3.1 Denominators and qualifiers

An isolated percentage has little structural meaning unless its denominator remains accessible. The statement 3 of 12 records (25%) is therefore one linked observation in this synthetic narrative. The fraction must not migrate to a caption or be attached to the following sentence.[5]

For the fixture, a positive label denotes only the presence of a label in the source. It does not establish a clinical outcome. Likewise, a negative statement must remain negative: no occlusion was recorded is not equivalent to occlusion was recorded. This contrast is included to expose dropped negation.

A qualified statement

The narrative says that the second observation was not confirmed. This is intentionally different from saying that it was disproved. A translation or a schema that allows only true and false would erase that distinction.

17.3.2 Traceable interpretation

Each interpretation should point to a source span, table cell or figure region. If two records disagree, both source values should remain available. Choosing the more plausible value without retaining the discrepancy would make the transformation difficult to audit.

The same principle applies to page references. Printed page 404 is physical PDF page 4 in this fixture. A reference to Table 17.1 should resolve to the logical table, while an instruction to inspect its second fragment should resolve to this physical page.

Synthetic endpoints

Terms such as primary patency, reintervention and follow-up appear here as terminology specimens. The chapter defines no treatment endpoint and reports no real outcome. Its purpose is to preserve the distinction between narrative, data and layout.[6]

TABLE 17.2 Numeric and lexical specimens

Source token	Meaning in this fixture
0.05 / 0.5 / 5.0	Three different decimal values
1,250 / 1.250	Separator must be preserved
≤ 0.80 / ≥ 1.20	Inequality direction matters
5×10^3	Exponent is three
37 °C / 12 µm	Units are part of the token
8-12 / follow-up	Range versus lexical hyphen
not present / present	Negation must survive
0 / NR / blank	Zero, not reported, missing

17.4 A DEFINED TEST INDEX

The following index is invented for the fixture. It is not a validated vascular score. Its purpose is to test subscripts, operators and a self-contained equation.

$$D_{test} = (Q_{out} / Q_{in}) \times 100\%$$

Here Q_{out} and Q_{in} are arbitrary nonzero scalars. If $Q_{out} = 3$ and $Q_{in} = 4$, the defined test value is 75%. This arithmetic example has no clinical interpretation.

Ordered review sequence

1. Identify the source block.
   a. Preserve its printed unit.
   b. Keep any qualifier.
2. Compare the extracted value.
3. Record a discrepancy explicitly.

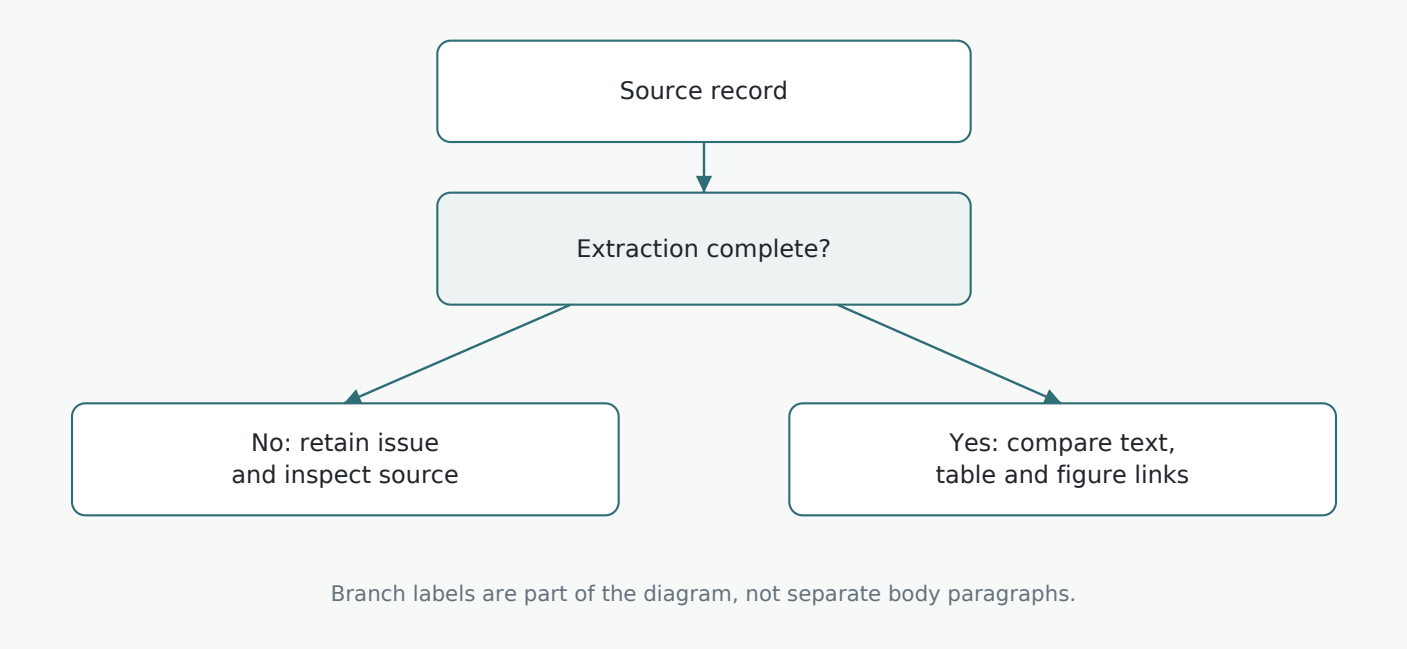


FIGURE 17.2 A synthetic review route. The artwork consists of vector shapes, native text and connecting arrows. A successful parser retains it as one figure region and excludes its labels from surrounding body prose.

17.5 REPORTING AND LIMITATIONS

A structured report should allow the reader to return from any extracted statement to its source. Coordinates alone are insufficient when a table continues onto another page; the logical identifier and the fragment identifier are both useful. For a scanned source, the preserved page image is the final visual reference.

The fixture separates a generated draft from an accepted interpretation. A later change should not overwrite the source observation or silently alter a previous version. The term reconstruction is used consistently in the glossary, but the abbreviation in the source remains available.

> **CASE VIGNETTE 17.1 - FICTIONAL**
> Record VX-104 has a left-side label and a right-side label in different source blocks. The discrepancy is unresolved.
> Expected handling: retain both labels and flag the conflict; do not choose a side automatically.

The final page is an image-only facsimile of a synthetic insert. It includes searchable-looking text that is actually raster content. Detecting the absence of native text is therefore part of the input classification task.[7,8]

REFERENCES - ALL SYNTHETIC

1. Fixture Editorial Group. Structured observations in an invented vascular chapter. *Synthetic Vascular Methods.* 2026;1:11-18. TEST-REF-001.

2. Parser Bench Collective. Separating page furniture from surgical narrative. *Mock Document Studies.* 2026;2:21-29. TEST-REF-002.

3. Illustration Test Unit. Composite figures and image-only labels. *Fictional Imaging Notes.* 2026;3:31-39. TEST-REF-003.

4. Table Fixture Group. Continued tables with merged cells. *Synthetic Layout Review.* 2026;4:41-48. TEST-REF-004.

5. Numeric Audit Team. Denominators, decimals and missing values. *Mock Data Reports.* 2026;5:51-59. TEST-REF-005.

6. Terminology Test Group. Context-bound glossary entries. *Synthetic Language Records.* 2026;6:61-67. TEST-REF-006.

7. OCR Fixture Unit. Native text versus page images. *Mock Extraction Research.* 2026;7:71-79. TEST-REF-007.

8. Version Test Group. Traceable correction of an invented record. *Fictional Audit Journal.* 2026;8:81-89. TEST-REF-008.

TABLE 17.3 Mixed-script glossary specimens

English source	Georgian terminology specimen
reconstruction	რეკონსტრუქცია
endovascular treatment	ენდოვასკულური მკურნალობა
inflow	შემომავალი ნაკადი
outflow	გამომავალი ნაკადი
not assessed	არ შეფასებულა
follow-up	შემდგომი დაკვირვება

Bibliography entries are invented and must not be treated as real literature. The glossary is a parsing specimen, not a comprehensive terminology standard.

17.6 IMAGE-ONLY SUPPLEMENT

SYNTHETIC SCANNED INSERT. This whole page is a raster image in the delivered PDF. It has no hidden searchable text layer. Its words, table and diagram require optical character recognition.

Supplemental observation

Specimen SCAN-201 retains the statement: no occlusion was recorded. The number 0.05 must remain distinct from 0.5. The side is L. Every observation on this page is fictional and supplied only for parser testing.

A parser should identify this page as requiring OCR while preserving the original image. Page furniture still remains separate from the main text. A failed OCR attempt is a processing issue, not evidence that the source page was empty.

Image and caption

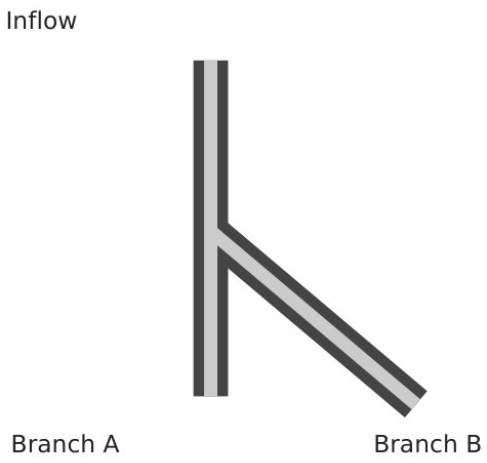


FIGURE 17.3 Original scanned diagram. The figure and its caption are both rasterized with the page.

TABLE 17.4 Image-only observation grid

Record	Side	Value	Qualifier
SCAN-201	L	0.05	not confirmed
SCAN-202	R	0.50	not assessed
SCAN-203	L	0	measured zero
SCAN-204	R	NR	not reported

SCAN CHECK. OCR should recover four data rows. No occlusion was recorded remains a negative statement. The page number is 407, while its physical PDF index is 7.